

Hit or Miss: Predicting the Success of Songs on Spotify



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Ng Chee Wei Nate

Agenda

- Problem statement
- From data gathering to feature selection
- Model training - process and workflow
- Final model
- Deployment demo - Streamlit

Problem Statement

- Do Spotify's top hits have common characteristics that set them apart from average songs?



Top Tracks of 2020

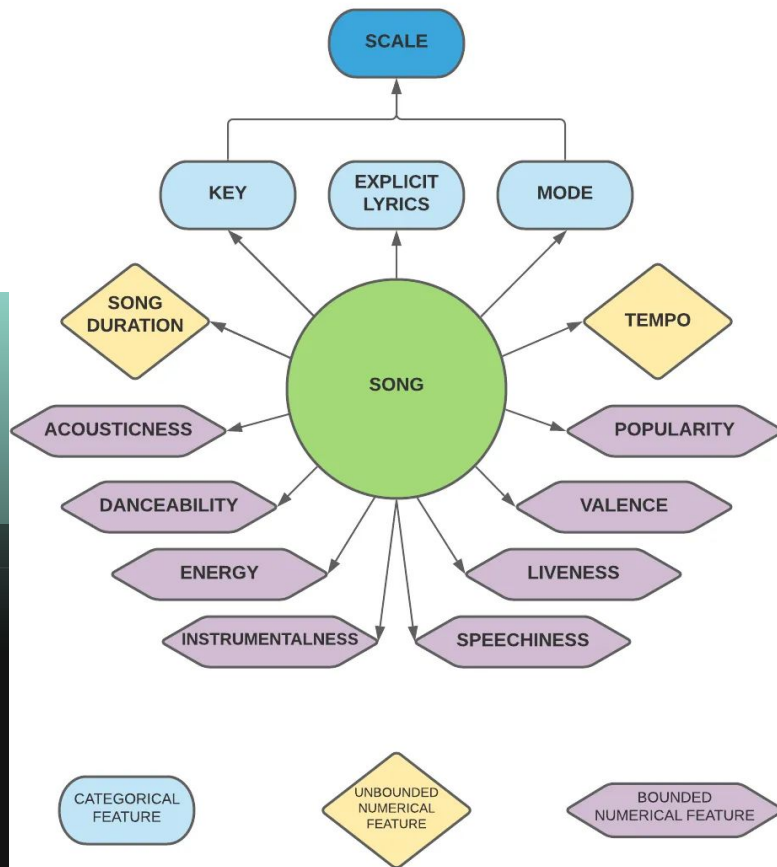
Playlist

Top Tracks of 2020

The Global Top Tracks of 2020. Cover: The Weeknd

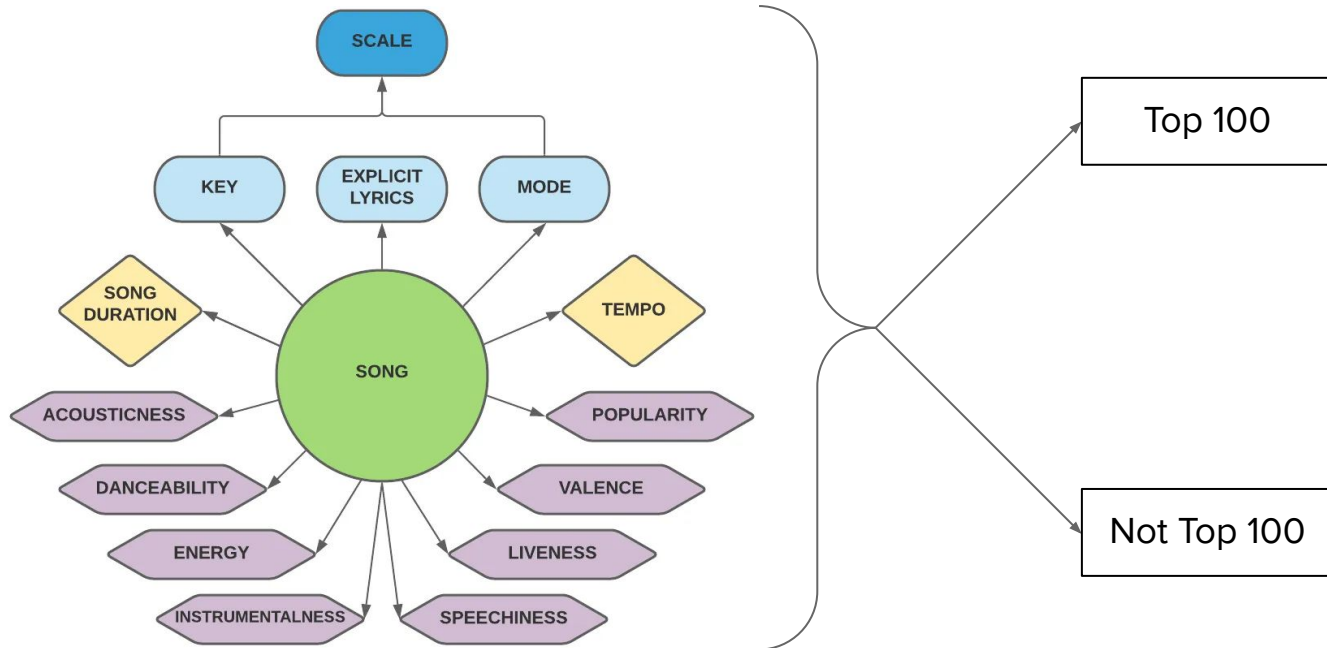
Spotify • 145,241 likes • 50 songs, about 2 hr 30 min

#	Title	Album	Date added	
1	 Blinding Lights The Weeknd	After Hours	1 Dec 2020	3:20
2	 Dance Monkey Tones And I	Welcome To The Madhouse (Deluxe)	1 Dec 2020	3:29
3	 The Box Roddy Ricch	Please Excuse Me for Being Antisocial	1 Dec 2020	3:16
4	 Roses - Imanbek Remix SAINT JHN, Imanbek	Roses (Imanbek Remix)	1 Dec 2020	2:56
5	 Don't Start Now Dua Lipa	Future Nostalgia	1 Dec 2020	3:03
6	 ROCKSTAR (feat. Roddy Ricch) DaBaby, Roddy Ricch	BLAME IT ON BABY	1 Dec 2020	3:01
7	 Watermelon Sugar Harry Styles	Fine Line	1 Dec 2020	2:54

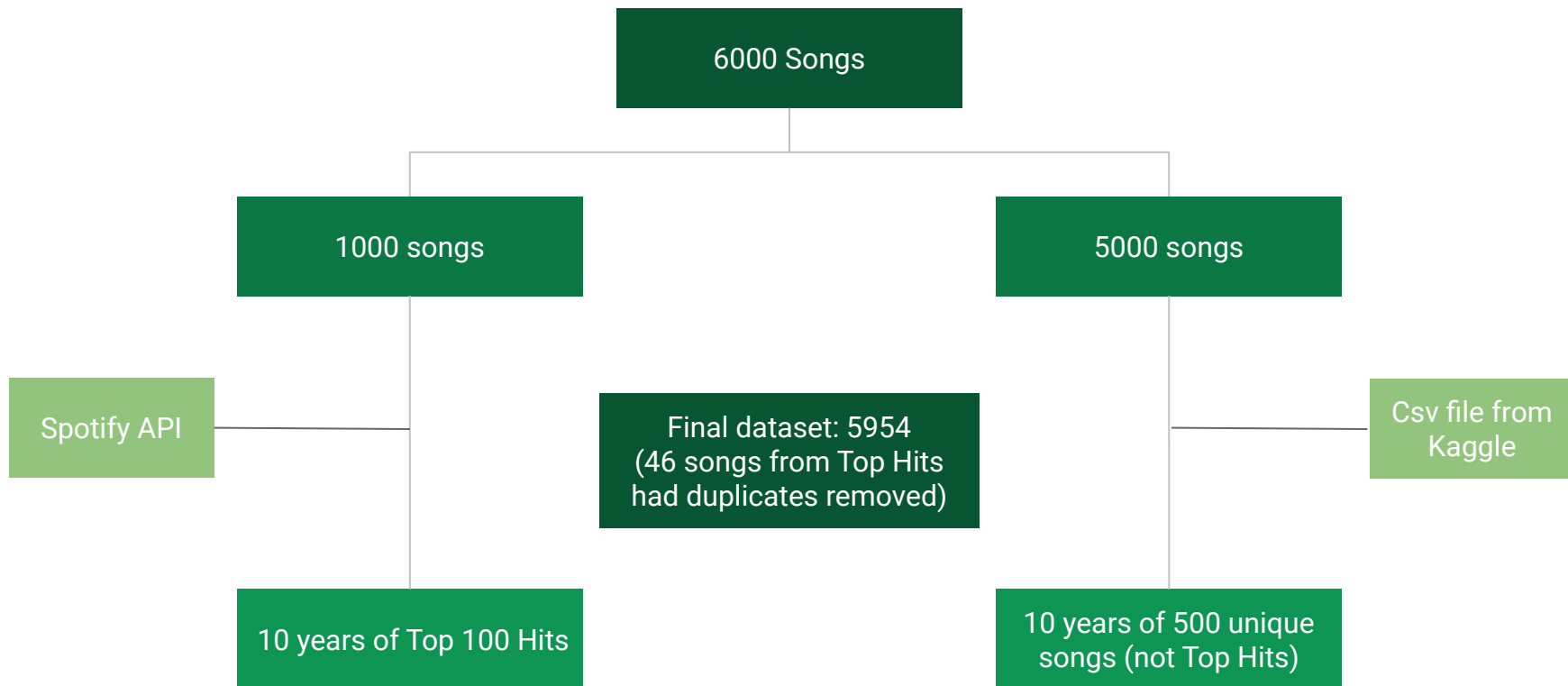


Problem Statement

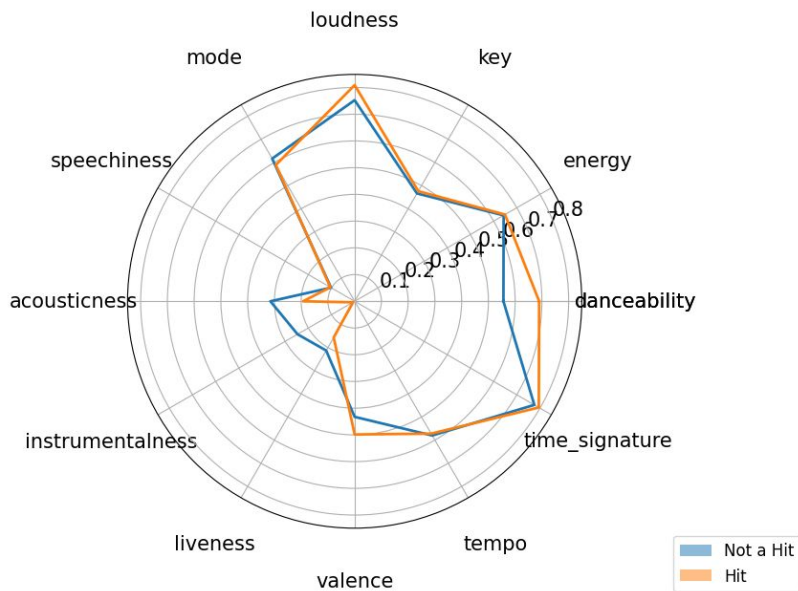
- Can we develop a classification model which predicts if a song will make it to the top 100, based on its audio features?



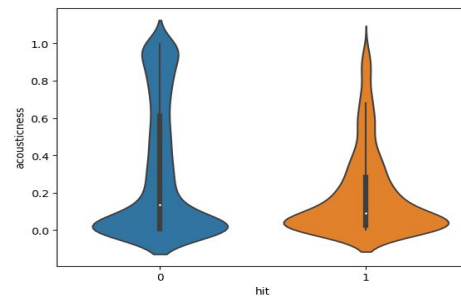
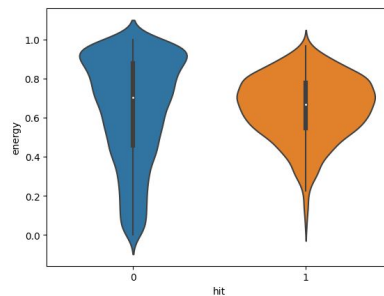
Data Collection



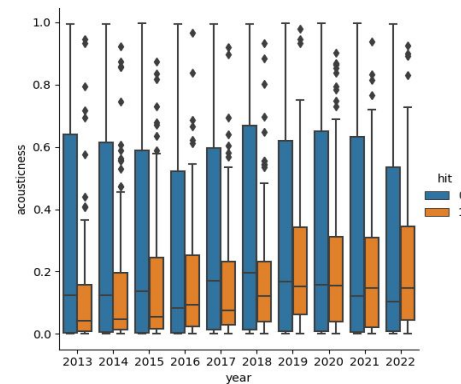
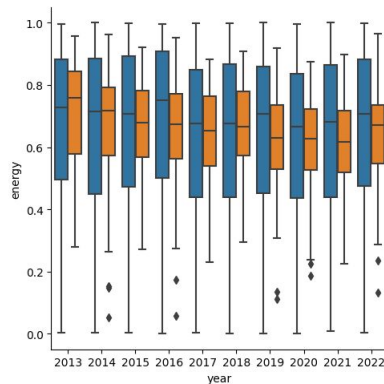
Data Observations



Differences in the mean of features



Differences in distribution



Whether trends changed with time

Evaluation Metric

F1

- Combination of precision and recall
- Generally, higher F1 (for minority class) is preferred
- When F1 scores of multiple models are really close, we look at precision

ROC-AUC

- AUC reflects the ability to classify between positive and negative class
- Higher is better

Experiments [Nate]

Trying a Variety of Models

Logistic Regression

Testing Thresholds

K Nearest Neighbors

Support Vector Machine

Random Forest

Balanced Random Forest

Ada Boost

XGB Boost

Decision Tree

Oversampling (Smote and Random & Undersampling

Logistic Regression

K Nearest Neighbors

Random Forest

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XGB Boost

Hyperparameter tuning (GridSearch)

Stacking and Voting Ensemble Classifiers

Random Forest

Ada Boost

XGB Boost

Additional data prep

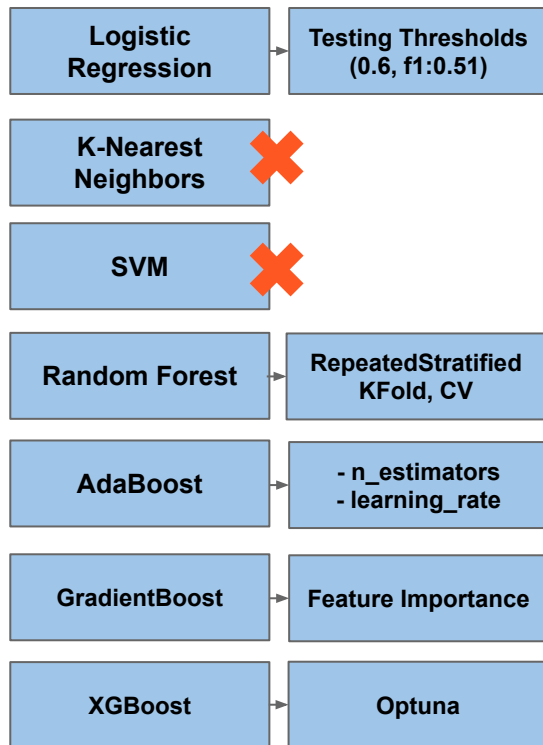
- One-hot Encoding
- Conversion of 'loudness'
- Feature Importance – 10 features vs 4 features
- Stratified split and scaling where necessary

Experiments [Louis]

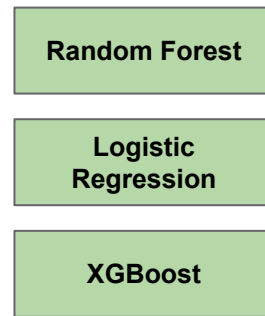
Additional data prep

- Minmax, standard scaler
- Label Encoder
- Dropping irrelevant columns ('track_name', 'artist_name')
- Gridsearch for best class weights

Trying a Variety of Models



Stacking and Voting Ensemble Classifiers



Smote and Random Undersampling



Overall Results

Random Oversampling,
Random Forest with Tuned
Hyperparameters
(ROC_AUC: 0.892)

	Prec.	Recall	F1
0	0.95	0.84	0.89
1	0.49	0.78	0.60

n_estimators=100, max_depth=10,
max_features=7, min_samples_leaf=1,
min_samples_split=3, random_state=42

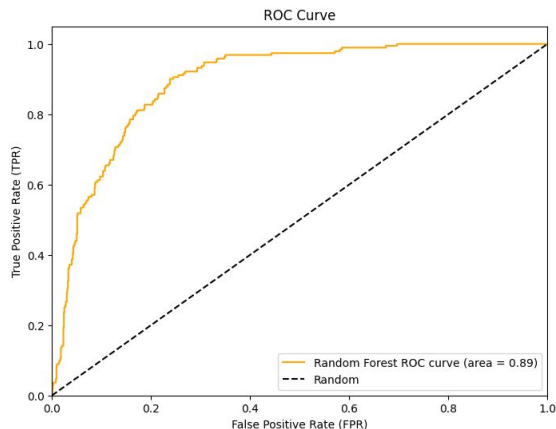
Voting Classifier (Hard)
Random Forest, AdaBoost,
XGB Boost
(ROC_AUC: 0.899)

	Prec.	Recall	F1
0	0.93	0.91	0.92
1	0.56	0.62	0.59

Voting Classifier (Soft)
Random Forest, Logistic
Regression, XGB Boost
(ROC_AUC: 0.787)

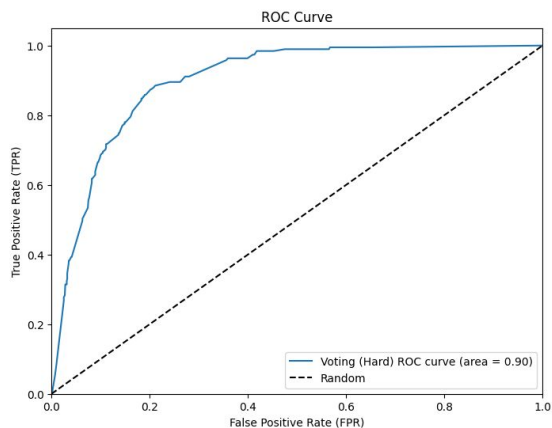
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0	0.94	0.87	0.90
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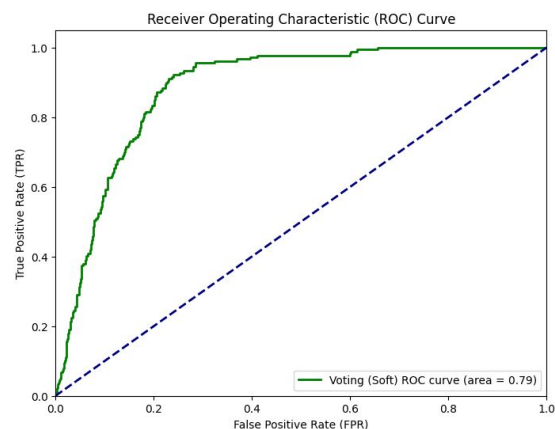
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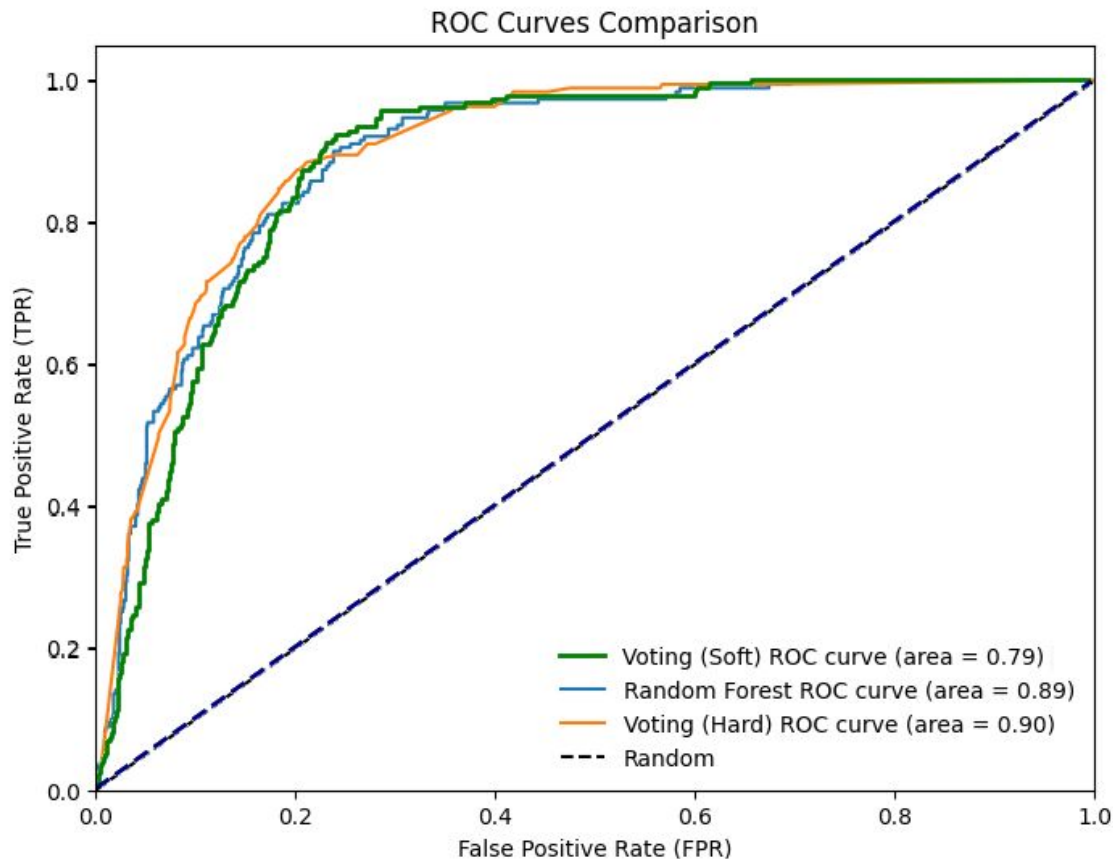
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Web Deployment

<https://spotifypredapp-cvmugwf2pxpmrztam4z2yn.streamlit.app/>

Final Feature Set

- Danceability
- Energy
- Linear Loudness
- Speechiness
- Acousticness
- Instrumentalness
- Liveness
- Tempo
- Valence
- Duration ms

HIT OR MISS?

Search results for cruel summer :

- 1 . Cruel Summer by Taylor Swift
- 2 . Cruel Summer by Bananarama
- 3 . Cruel Summer by Taylor Swift
- 4 . Cruel Summer by Kidz Bop Kids
- 5 . cruel summer by The Maine
- 6 . Cruel Summer (Sped Up) - And It's You, the Shape of Your Body by Hiko
- 7 . Cruel Summer by Bananarama
- 8 . cruel summer - piano instrumental by Chilled Pig
- 9 . Cruel Summer - Sped Up by Speed Sounds
- 10 . Cruel Summer (Marcus Layton Edit) by LAUWE

Select a song number:

1

Predict if this song is a hit!

How could we improve the prediction?

- **Include** Artist popularity
 - Possible that the new songs from popular artists are more likely to top the charts
 - BUT: this will change over time i.e. an artist popular now may not be popular 10 years on
 - Disadvantages a new artist
- **Include** Followers
 - Another way to track artist popularity
 - Effectiveness is time sensitive: numbers change over time
- Work on smaller segments
 - 1 year instead of 10 years (2013 only)
 - 4 years 2019-2022
 - Results were not better

popularity integer

The popularity of the artist. The value will be between 0 and 100, with 100 being the most popular. The artist's popularity is calculated from the popularity of all the artist's tracks.

followers object

Information about the followers of the artist.